Skills Summary

Bounds, Signs, and the Variable of Integration

Use this reference while completing your worksheet or when studying.

Skill 1 — Limits: Order and Conversion

Order. $\int_b^a f = -\int_a^b f$. A definite integral written with the larger number on the bottom is not a typo to be tidied away — it is the negative of the usual one.

Conversion. After a substitution, the limits belong to the new variable. Convert them at the same moment you convert the integrand.

Example: Evaluate $\int_4^1 2x\,dx$.

Step 1. The lower limit is larger than the upper one, so the value will be negative; evaluate in the order written rather than swapping the numbers.

Step 2. $[x^2]_4^1 = 1 - 16$

$\Rightarrow$ $\mathbf{-15}$

Try it: Evaluate $\int_3^0 2x\,dx$.

check: $\mathbf{-9}$

Skill 2 — Signed Integral or Geometric Area

Two different questions. $\int_a^b f$ counts area below the axis as negative. *Area* counts it as positive, so split the interval at every sign change and add the magnitudes.

The same distinction separates displacement (signed) from total distance travelled (unsigned) in a motion problem.

Example: Find the area between $y = x$ and the x-axis on $[-1, 1]$.

Step 1. The question asks for area, and the graph crosses the axis inside the interval, so the signed integral would cancel to zero and must not be used.

Step 2. Split at 0: each triangle has area $\frac{1}{2}$, so the total is $\frac{1}{2} + \frac{1}{2}$.

$\Rightarrow$ $\mathbf{1}$

Try it: Find the area between $y = x$ and the x-axis on $[-2, 2]$.

check: $\mathbf{4}$

Skill 3 — Substitution Bookkeeping

Three things change together. The integrand, the differential, and the limits. If $u = g(x)$ then $du = g'(x)\,dx$ — solve that for the piece you actually have, and carry any constant factor out front.

An answer that still contains the old variable, or that uses the old limits, is half-converted.

Example: Evaluate $\displaystyle\int_0^1 x\left(x^2+1\right)^2 dx$.

Step 1. The integrand contains a function and a multiple of its derivative, so substitution is the tool; the limits must move with it.

Step 2. $u = x^2+1$, $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$ and u runs from 1 to 2: $\frac{1}{2}\left[\frac{u^3}{3}\right]_1^2 = \frac{1}{2}\left(\frac{8}{3}-\frac{1}{3}\right)$.

$\Rightarrow \quad \boldsymbol{\frac{7}{6}}$

Try it: Evaluate $\displaystyle\int_0^1 x\left(x^2+1\right)^3 dx$.

check: $\boldsymbol{\frac{8}{15}}$